

Original Article

Genome-wide identification, expression profiling, and network analysis of AT-hook gene family in maize

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ABSTRACT

AT-hook motif nuclear localized (AHL) genes have diverse but poorly understood biological functions. We identified and analyzed 37 *AHL* genes in maize. We also discovered four and one additional AHLs in rice and sorghum, respectively, besides those reported earlier. The maize AHLs were classified into two clades (A and B) and three distinct types (I, II, and III) as also reported in *Arabidopsis*. Phylogenetic and ortholog analyses showed that, while the evolutionary classification was conserved in plants, expansion of the *AHL* gene family in maize was accompanied with new biological functions. Gene structure analysis showed that, while all but one Type-I AHLs lacked an intron, origin of Type-II and Type-III AHLs was associated with the gain of introns suggesting evolutionarily distinct temporal and spatial expression patterns and, likely, neofunctionalization. Gene duplication analysis revealed that AHLs in maize expanded via dispersive duplication further supporting their functional diversity. To discern these functions, we analyzed 71 transcriptomes from diverse tissues and developmental stages of maize and classified AHLs into eight groups with distinct temporal/spatial expression profiles. Coexpression analysis implicated 5 AHLs and 33 novel genes in networks specific to endosperm, seed, root, leaf, and reproductive tissues indicating their role in the development of these organs. Major processes coregulated by AHLs include pollen development, drought response, senescence, and wound response. We also identified interactions of AHL proteins in coregulating important processes including stress response. These novel insights into the role of AHLs in plant development provide a platform for functional analyses in maize and related grasses.

1. Introduction

The AT-hook motif nuclear localized (AHL) gene family is highly conserved in all land plants. While such conservation indicates the biological importance of this gene family, the precise role of AHLs in plant growth and development is only beginning to be understood. Like the SQUAMOSA-promoter binding like (SPL) gene family in barley (*Hordeum vulgare* L.) [1], members of the *AHL* gene family have been shown to regulate diverse developmental processes including flower development [2], hypocotyl elongation [3,4], pollen development and fertility [5,6], development of maize (*Zea mays* L.) ears [7], differentiation of vascular tissues [8], regulation of flowering time [9–11], and regulation of leaf senescence [12]. AHLs have also been implicated in plant responses to biotic and abiotic stresses including drought [13,14] and pathogens [15,16]. Other major functions of AHLs include

regulation of primary metabolism [17], homeostasis of gibberellins [18], and regulation of jasmonic acid and auxin-related gene expression [13,19,20]. Finally, by regulating epigenetic states, AHLs perform diverse functions including preservation of genome integrity by epigenetic silencing of transposable elements [9], and preventing interploidy hybridization and thus playing a key role in speciation [21]. Despite such wide-ranging roles, a comprehensive understanding of the sequence and functional diversity of the *AHL* gene family is lacking.

Based on phylogenetic analysis of these proteins in land plants, AHLs have been divided into two distinct clades referred to as Clade-A and Clade-B [4,18,22,23]. The AHL proteins are characterized by the presence of an AT-hook motif, and a Plant and Prokaryotic Conserved (PPC) domain also referred to as Domain of Unknown Function (DUF) 296 [4]. The AT-hook motif was first described in high mobility group (HMG) proteins involved in orchestrating the replication,

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recombination, transcription, and DNA repair [24,25]. Several other eukaryotic proteins including histones, homeodomain, PHD, and Zinc finger proteins have since been found to carry the AT-hook motif [26]. This motif has a characteristic core sequence of Arg-Gly-Arg amino acids (aa) with either side flanked by Arg-Lys or Pro residues. The AT-hook motif uses Arg-Gly-Arg residues to bind with AT-rich regions (AA(T/A)T) of DNA, forming a concave conformation to firmly insert R residues in the minor groove [27]. The AT-hook motifs also have consensus variation resulting in their divergence into Type-I motifs with a consensus sequence of Gly-Ser-Lys-Asn-Lys and Type-II motifs with Arg-Lys-Tyr at the C-terminus side of the Arg-Gly-Arg core. The second structural unit, the PPC domain, contains around 120 aa [22]. This conserved domain, which appears on the carboxyl end of the AHL protein, is vital in the nuclear localization of AHLs for their DNA binding [4]. Based on sequence variation, the PPC domain has been classified into two types: Type-A domain contains a conserved core sequence of Leu-Arg-Ser-His while Type-B domain consists of Phe-Thr-Pro-His in upstream regions [23]. The PPC domain sometimes carries Gly-Arg-Phe-Glu-Ile-Leu sequence shown to be essential for the functionality of some of the AHL proteins by mediating interaction with transcription factors [4]. Based on the abundance and composition of the two structural units, AT-hook motif and PPC domain, AHL proteins are classified into three subtypes. The Type-I AHLs contain one Type-I AT-hook motif and a Type-A PPC domain. The Type-II AHLs contain two AT-hook motifs (Type-I and Type-II) chained together along with a Type-B PPC domain. The Type-III AHLs contain one Type-II AT-hook motif and one Type-B PPC domain. Finally, Type-I AHLs belong to Clade-A while Type-II and Type-III AHLs are members of Clade B [23].

Through genome scans targeting the two characteristic functional units, the AT-hook motif and the PPC domain, *Arabidopsis thaliana*, rice (*Oryza sativa* L.), and sorghum (*Sorghum bicolor* L.) have been reported to contain 29, 24 and 22 AHLs, respectively [18,23,28]. However, the functional analysis of a small number of AHLs has been mostly limited to *Arabidopsis*. Maize, an important cereal, has emerged as a model crop plant for studying genetics, evolution, and other biological processes. Maize productivity is continually affected by biotic and abiotic stresses. While AHLs are expected to play very important and diverse roles in maize, a detailed genome-wide analysis of AHL gene family has not been performed. The maize genome was sequenced a decade ago and underwent three rounds of revisions [29]. Recently an entirely new assembly (B73_RefGen_v4) was constructed based on PacBio single-molecule real-time sequencing and high-resolution optical mapping [30]. The newer version, which has a 52-fold larger contig length and other improvements over the previous version, provides a new foundation for genomic interrogation. In earlier studies, we have constructed over 79 transcriptomes of distinct tissues representing distinct organs and developmental stages [31–33]. The availability of these genomic and transcriptomic resources provides an excellent opportunity for annotation of AHLs, their functional classification based on temporal and spatial expression patterns, and identification of AHL-interacting genes through coexpression network analyses. Given their diverse roles in an array of biological processes, characterization of AHLs in maize will enhance our fundamental understanding about this gene family and allow the application of such knowledge for the genetic improvement of maize. A similar characterization of many gene families including WRKY, MADS, and NAC [34,35] has facilitated the identification of important target genes for improvement of yield, stress tolerance, and other traits [36–38]. The objective of this study was comprehensive evolutionary and functional analysis of AHL gene family in the model crop maize and comparative genomic analysis in three major plant species. This study enhances the understanding of diversity and novel interactions of the AHL genes in plant development and paves the way for targeting this gene family for improvement of productivity and stress tolerance in grasses.

2. Materials and methods

2.1. Genome-wide identification of AHL proteins

For the discovery of the AHL family members, latest release of the reference maize genome B73_RefGen_v4 (Zea_mays.AGPv4.40.gtf) was retrieved from Ensembl Plants (https://plants.ensembl.org/Zea_mays/Info/Index) [39]. The Hidden Markov Model (HMMscan) scan was used on the complete maize proteome using PPC/DUF296 (PF03479) Pfam ID obtained from the Pfam database (<https://pfam.xfam.org/>) [40]. The presence of characteristic domains in the retrieved sequences was confirmed by searching against the Conserved Domains Database (www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi) using e-value of 10^{-5} . The unique sequences containing the PPC domain and the AT-hook motif were selected as putative AHLs. These sequences were annotated by searching against the NCBI non-redundant database (e-value 10^{-5}). The gene ontology mapping was carried out using Maize Mine v1.1 (<http://maizemine.rnet.missouri.edu:8080/maizemine/begin.do>) available at MaizeGDB [41].

2.2. Genes structure and in-silico characterization

The motifs in AHL protein sequences were predicted by MEME with e-value 10^{-5} [42]. Exon and intron arrangements were determined by aligning the coding sequences with genomic sequences obtained from Ensembl Plants (<http://plants.ensembl.org/index.html>). The effect of introns on the enhancement of gene expression was analyzed using IMter server (<http://korflab.ucdavis.edu/cgi-bin/web-imeter2.pl>) [43]. An IMter score of 0 to 100 was used to analyze the effect on gene expression, and a score > 20 was considered to have a high effect on expression enhancement [43,44]. The molecular weight (MW), isoelectric point (pI), sub-cellular localization, and signal peptides were predicted using ExPasy MW/pI (https://web.expasy.org/compute_pi/) [45], CELLO v.2.5 (<http://cello.life.nctu.edu.tw/>) [46], and SignalP 5.0 Server (<http://www.cbs.dtu.dk/services/SignalP/>) [47], respectively.

2.3. Prediction of orthologs

A Pfam domain PPC/DUF296 (PF03479) was used as a query in Ensembl Plants (<http://plants.ensembl.org/index.html>) Biomart tool [48] to identify the AHL proteins of sorghum, rice, and *Arabidopsis*. The retrieved sequences were annotated by NCBI CDD (www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi). The orthologous genes of maize AHLs were predicted using reciprocal blast (e-value 10^{-5}) against *Arabidopsis*, rice, and sorghum proteins.

2.4. Phylogenetic analysis

A Neighbor-Joining tree was constructed for inferring the phylogenetic relationship among the 37 maize AHL proteins and with *Arabidopsis*, rice, and sorghum AHL proteins [49]. The AHL proteins sequences were aligned using ClustalW in MEGA7 [50]. The 1000 Bootstrap replicates were applied to prepare this consensus to represent the evolutionary history [51].

2.5. Gene expression analysis

Gene expression data for different development stages of a number of organs including leaf, root, seed, embryo, endosperm, internodes, and reproductive tissues (tassel, silk, and cob) of B73 inbred line [32,33] was used for expression-based classification of AHLs. Since the previously published RNA-seq data is based on an older version of maize genome, the RNA-seq reads were aligned to the current reference maize genome (Zea_mays.AGPv4.40.gtf) [30] with the Tophat2 package (v. 2.1.1) [52] and Bowtie2 (v. 2.3.4.1) short-read aligner [53] followed by calculation of fragments per kilobase million (FPKM) with

Cufflinks v.2.2.1 [52]. The log2 transformed FPKM values were used for preparing the heatmaps.

2.6. Coexpression networks

Coexpression analysis was performed using the FPKM data set described above for seed, endosperm, embryo, leaf, and reproductive tissues [32,33]. Coexpression networks were constructed for each of the five organs/organ groups listed above using random matrix theory (RMT) based approach [54]. For each group, genes with FPKM < 1 in > 20% of samples were removed from the analysis. The AT-hook binding sites (AA(T/A)T) in the promoter region of genes coexpressed with the AHL genes were predicted using PlantCARE database [55].

2.7. AHL proteins interaction network analysis

The physical interactions of AHL proteins with other cellular proteins were determined using STRING v11.0 (<https://string-db.org/>) [56]. The 37 maize AHL protein sequences were used as query against the maize data in the STRING database. The analysis was performed by keeping a high level of confidence (0.7) with a maximum of 20 interaction in first and second shells.

2.8. Gene duplication events

MCSanX [57] was used to compute the duplications events for the maize AHL genes. The AHL proteins were used as query in BLASTp against maize protein database with e value-10⁻⁵. The output with top five protein hits was used in duplicate_gene_classifier function of MCSanX to classify whole genome/segmental, tandem, proximal and dispersed duplications.

3. Results and discussion

3.1. Genome-wide analysis reveals diversification of AHLs in maize

We identified 37 AHL protein sequences in the maize genome that carried both the PPC domain and the AT-hook motif (Fig. 1, Supplementary Table S1). In the maize AHLs, AT-hook motifs range from 1 to 24 and the PPC domains range from 1 to 5. The Gly-Arg-Phe-Glu-Ile-Leu residues found in the PPC domain of 18 (62%) of *Arabidopsis* AHLs [4] were present in 19 (51%) maize, 12 (48%) rice, and 12 (46%) sorghum AHLs. The presence of these functional residues in nearly half of the AHL proteins identified in diverse plant species supports a conserved function of AHLs during evolution and divergence. Divergence of the PPC domain in the rest of the AHLs could allow flexibility in protein complex formation thereby expanding the biological roles of AHLs.

To infer the evolutionary relationship among the AHL proteins and to classify them based on the presence of characteristics sequence motifs, phylogenetic analysis was performed using the full-length protein sequences. This analysis showed the diversification of AHL proteins into two clades, Clade-A consists of 16 proteins and Clade-B contains 21 proteins (Fig. 2). A similar classification of AHLs into two clades has been reported for *Arabidopsis* and rice [18,28]. Upon multiple sequence alignment and examination of the presence and combination of the two characteristic functional units, the PPC domain and the AT-hook motif, the AHLs were further classified into Type-I, Type-II and, Type-III (Fig. 2). Of 37 AHLs, 21 (57%) belong to Type-I, 10 (27%) are Type-II, and remaining 6 (16%) correspond to Type-III. The Type-I AHLs belong to Clade-A while Type-II and Type-III to Clade-B. Remarkably, the fraction of genes in each of the three AHL types and the two clades in maize is similar to that observed in other plants with the highest number of genes belonging to Type-I (Clade-A) and lower number classified as Type-II and Type-III (Clade-B) [23]. This observation

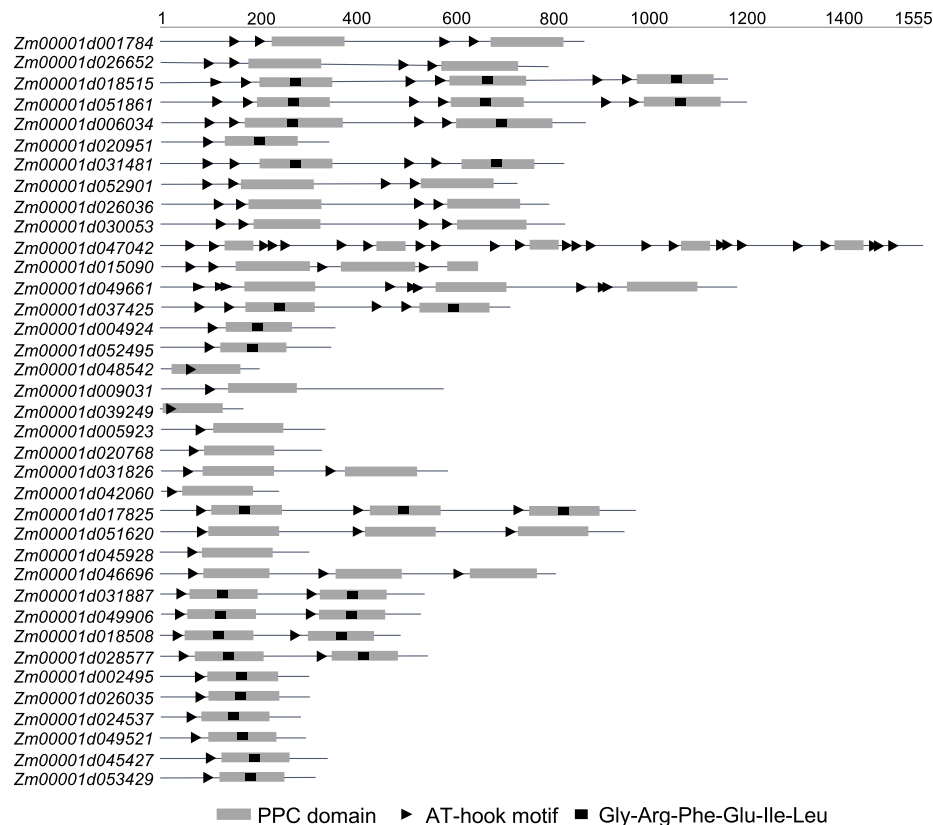


Fig. 1. Domain structure and motif analysis for characterization of AHL proteins. Solid lines denote the protein length and various shapes represent key features of the proteins. Scale at the top denotes protein size.

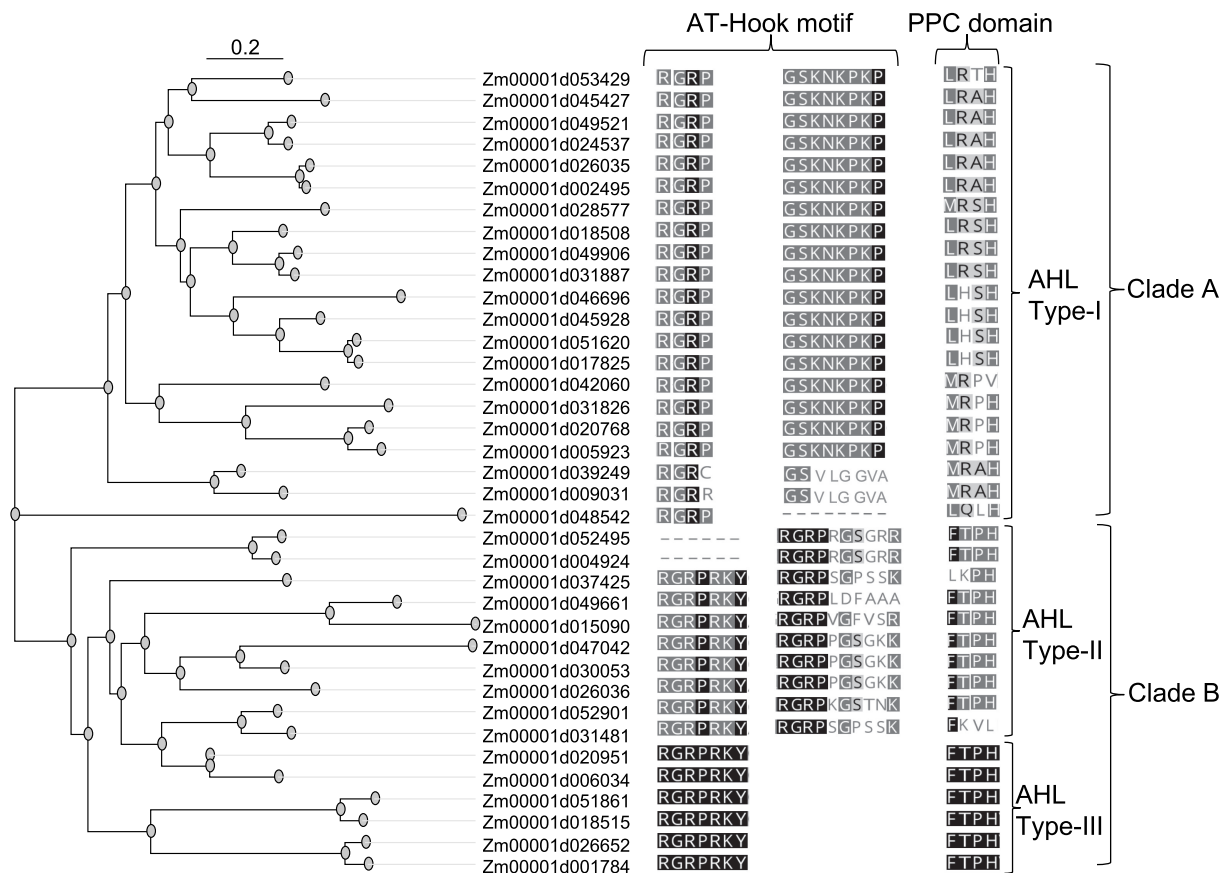


Fig. 2. Phylogenetic analysis of the maize AHL proteins. Phylogenetic tree is shown on the left and characteristic sequences in two the functional units of the AHL proteins, AT-hook motif and PPC domain, are shown on the right.

indicates that the evolutionary diversification of AHLs is conserved across monocots and eudicots.

The Type-I AHLs contain AT-hook motifs with conserved sequences of Arg-Gly-Arg-Pro followed by Gly-Ser-Lys-Asn-Lys on the C-terminus side and a PPC domain Type-A starting with less conserved sequence Leu-Arg-Ser-His on the N-terminus side (Fig. 2). Intriguingly, one of the Type-I AHLs (*Zm00001d048542*) has diverged from other sequences due to less conserved motifs associated with AHL proteins suggesting neofunctionalization. The Type-II and Type-III AHLs consist of Arg-Gly-Arg (RGR) AT-hook motif followed by Pro-Arg-Lys-Tyr on C-terminus side and Type-II AHLs have one additional AT-hook motif followed by Pro-Gly-Ser-Gly. Both Type-II and Type-III have a PPC domain Type-B starting with a less conserved sequence Phe-Thr-Pro-His on the N-terminus side. This classification of AHLs in maize is consistent with the proposed evolution of AHLs wherein the PPC domain was present in archaic AHLs, and the addition of one or more AT-hook motifs diversified AHLs in flowering plants [23]. However, lack of conservation of the sequence following AT-hook motif Type-II and the sequence of PPC domain Type-A, which are highly conserved in *Arabidopsis*, indicates functional diversification of AHLs in maize.

3.2. The gain of introns during AHL gene diversification suggests intron-mediated expression modifications in flowering plants

Gene structure analysis showed the presence of a variable number of introns and exons in the *AHL* genes (Fig. 3). All Type-I AHLs except one (*Zm00001d030053*) lack introns, while all Type-II and Type-III AHLs possess introns ranging between 2 and 5. All the prokaryotic genomes lack introns, and eukaryotes gained introns during or after prokaryote-

eukaryote divergence [58,59]. Therefore, Type-I AHLs are proposed to be the ancestral AHLs from which the Type-II and Type-III AHLs have evolved. Evolution of introns have been associated with alternative splicing and the resulting functional diversity in the encoded proteins [1,60–62], enhancement of gene expression [63], and nonsense-mediated decay of mRNA transport [64]. Diversification of AHLs in this manner confers functional evolution through duplication and subsequent differentiation.

To understand the biological role of introns sequences in AHLs, we examined the capabilities of these introns to enhance the transcription of the associated genes. The first intron of two of the AHL Type-III genes, *Zm00001d018515* and *Zm00001d051861*, was predicted to enhance the expression of these genes (Supplementary Table S2). Remarkably, in agreement with this prediction, both these genes are constitutively expressed in all the tissues analyzed (see below in Fig. 5). To summarize, the gain of introns in Type-II and Type-III suggests their role in processes related to enhanced biological complexity in flowering plants. Finally, the discovery of one intron containing Type-I AHL (*Zm00001d048552*) seems to be an evolutionary link between Clade-A (Type-I) and Clade-B (Type-II and Type-III). The grouping of *Zm00001d048552* between Type-I and Type-II genes based on phylogeny (Fig. 2) also supports the idea of this gene being an evolutionary link between Type-I and Type-II AHLs. Since the addition of introns to genes is advantageous for the enhanced gene expression and regulation, origin of *Zm00001d048552* might have been associated with gain of a distinct function. Such neofunctionalization could have provided selective advantage to the modern plant species thus forming a basis for further expansion of the intron-carrying AHLs.

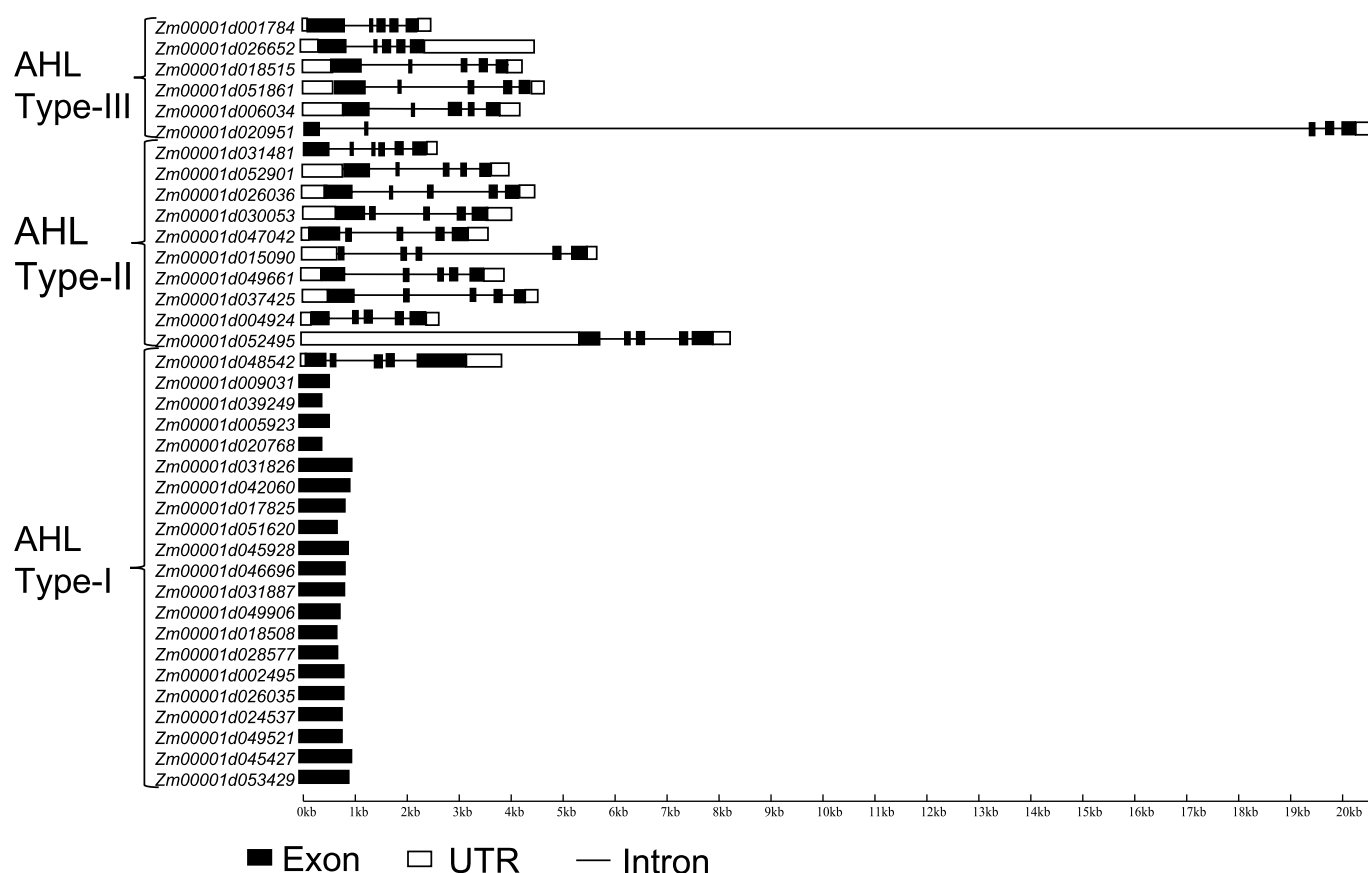


Fig. 3. AHL genes structure analysis. The exons and introns are represented as black boxes and thin black lines, respectively. UTRs are represented as open boxes. Scale for gene length is shown at the bottom.

3.3. Gene duplication analysis

The gene duplication in plants occurred through different mechanisms including whole genome/segmental, tandem, proximal, and dispersed duplications events. Herein, we analyzed maize AHLs for the presence of these duplication events. Remarkably, the analysis showed only the presence of 37 dispersed duplications events for AHLs in the maize genome that have contributed to expansion of this gene family. Unlike other duplication events, dispersed duplicated genes do not lie in the neighborhood of each other and are rather dispersed throughout the genome and associated with different regulatory elements [65]. Diversity of the regulatory elements would allow the regulation of different aspects of plant development. Consistent with the functional diversity of AHLs, dispersed duplication has contributed to their divergent gene expression and novelty in their functions in maize as also reported in *Arabidopsis* and rice [66]. Such diversity in the expression of duplicated genes is reflected in temporal and spatial expression pattern of AHLs in maize (Fig. 5). This analysis suggests novel role of AHLs in maize compared to those reported on other species and warrants further functional analysis.

3.4. Properties of the predicted maize AHL proteins

Based on in silico predictions [30], the Zm00001d048542 (573 aa) and Zm00001d039249 (155 aa) are the longest and shortest AHL proteins, respectively, and the average protein length is 336 aa (Supplementary Table S3). The predicted average molecular weight and isoelectric point (pI) range of AHLs is 34.02 and 5.08–10.28, respectively. Remarkably, a predicted signal peptide of 70 aa length is present in all the AHLs. Functional analysis of this conserved peptide sequence could provide insight into the regulatory role of the AHLs.

Most of the AHLs are predicted to be localized in the nucleus (33 proteins), followed by chloroplast (2 protein), plasma membrane (1 protein) and extracellular space (1 protein). The finding that the majority of AHLs are nuclear localized is consistent with the binding of these proteins to AT-rich regions of DNA [12] for performing regulatory functions. Multiple rice AHLs carrying a GFP tag were also exclusively detected in the nucleus [28]. GO enrichment analysis of maize AHLs also showed significant enrichment of DNA binding (9.180E-06) and nucleic acid binding (5.589E-04) as highly enriched further supporting the binding of AHLs to DNA for their regulatory functions. Predicted localization of a small number of AHLs in chloroplast raises an interesting possibility of AHLs performing their regulatory role by binding to chloroplast DNA.

3.5. Analysis of evolutionary relationships confirms divergence of AHLs in plants

To further examine the extent of divergence of AHLs in maize relative to other plants, we first annotated 29, 25, and 26 AHLs in *Arabidopsis*, rice, and sorghum, respectively. While a previous study [18] have reported the same number of AHLs in *Arabidopsis*, we identified one additional protein in sorghum and four additional proteins in rice compared to those reported earlier [23,28]. Notably, we found a Type-III AHL in sorghum while no members of this group appeared in an earlier study [23]. The identification of these additional AHLs in rice and sorghum showed that improvement in genome annotations can lead to characterization of useful and novel gene family members. Phylogenetic analysis among AHLs in these species showed a clear demarcation between Clade-A and Clade-B (Fig. 4). This observation is consistent with previous studies [4,18,22,23] and supports the idea that these clades are separated by a long evolutionary period. Clade-A

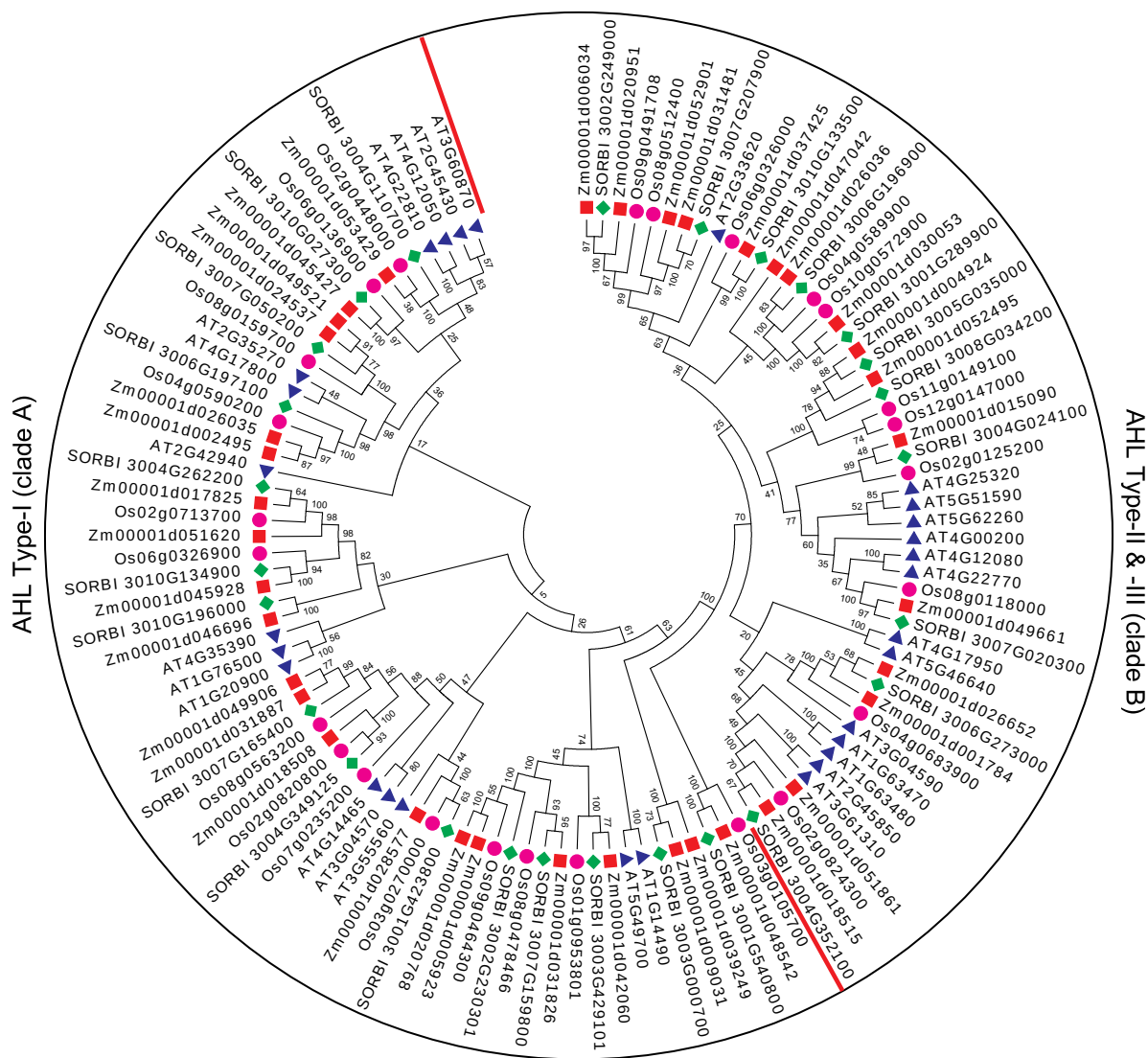


Fig. 4. Phylogenetic tree of AHL ortholog proteins. Combined phylogenetic tree for maize (red squares), *Arabidopsis* (blue triangles), rice (magenta ovals), and sorghum (green diamond) is shown. Solid red lines indicate the separation of AHLs belonging to Clade-A and Clade-B. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

represents primitive Type-I AHLs while Clade-B evolved during or after separation of land plants and mosses [4]. Lack of variation in Type-I AHL could also be due to a more fundamental function of these proteins resulting in a stringent purifying selection compared to the two other types. Finally, Type-II and Type-III AHLs did not form distinct subgroups within Clade-B suggesting a relatively recent divergence of these proteins.

Due to the functional conservation of proteins across species, determination of orthologous relationships can provide useful insights about the biological role of these proteins. To begin to understand the function of maize AHLs, we identified 11, 22, and 25 orthologs of maize in *Arabidopsis*, rice, and sorghum, respectively (Table 1). For two maize AHLs (Zm00001d026035 and Zm00001d024537), no orthologs were detected in any of the three species suggesting a maize-specific function for these proteins. Lack of orthology for all maize AHLs with those in the three species further indicates the divergence of AHLs, especially between monocots and eudicots. To the best of our knowledge, the only characterized AHL in maize is encoded by *barren stalk fastigate1* (Zm00001d045427) that controls ear development [7]. Due to lack of functional characterization in maize, putative maize orthologs of genes with a known function in *Arabidopsis* were designated accordingly

(Table 1). Remarkably, putative orthologs of maize AHLs are involved in diverse functions including drought tolerance, pathogen defense, senescence, hypocotyl elongation, and gibberellin homeostasis in *Arabidopsis* (Table 1). This observation highlights need for functional validations of the fundamentally and economically important perceived roles of AHLs in one of the most important food crops and an important crop model.

3.6. Gene expression patterns provide key insights into biological functions of AHLs in maize

As noted above, while substantial information has been generated on the role of AHLs in the model plant *Arabidopsis*, lack of information about their role in grasses is quite striking. To obtain clues about the role of these genes in maize development, we examined temporal and spatial expression patterns of the AHLs in 71 distinct transcriptomes in maize. RNA-seq data for distinct developmental stages of major organs including leaf (20), stem (10), root (11), reproductive (silk, anthers, cob, tassel) organs (6), embryo (5), endosperm (7), and seed (12) was interrogated. The expression of AHL genes showed remarkable variation and fell into three specific patterns that included constitutive

Table 1

The orthologs of maize AHLs in different species and their known functions.

#	Maize gene ID	Gene name	<i>Arabidopsis</i> gene ID	Gene name	Rice gene ID	Sorghum gene ID	Function (if known)
1	Zm00001d004924	–	AT2G36560	–	Os11t0149100	–	–
2	Zm00001d015090	ZmAHL1	AT4G12080	AHL1	–	SORBI_3004G024100	–
3	Zm00001d049661	ZmAHL1	AT4G12080	AHL1	Os08t0118000	SORBI_3007G020300	–
4	Zm00001d052495	–	AT2G36560	–	Os12t0147000	SORBI_3005G035000	–
5	Zm00001d031481	ZmAHL10	AT2G33620	AHL10	Os08t0512400	SORBI_3007G207900	Drought tolerance ^c [13]
6	Zm00001d037425	ZmAHL10	AT2G33620	AHL10	Os06t0326000	SORBI_3010G133500	Drought tolerance ^c [13]
7	Zm00001d052901	ZmAHL10	AT2G33620	AHL10	Os08t0512400	–	Drought tolerance ^c [13]
8	Zm00001d026036	ZmAHL13	AT4G17950	AHL13	Os04t0589900	SORBI_3006G196900	–
9	Zm00001d030053	ZmAHL8	AT5G46640	AHL8	Os10t0572900	SORBI_3001G289900	Hypocotyl growth ^c [4]
10	Zm00001d047042	ZmAHL13	AT4G17950	AHL13	Os10t0572900	–	–
11	Zm00001d048542	–	–	–	Os03t0105700	SORBI_3001G540800	–
12	Zm00001d005923	ZmAHL17	AT5G49700	AHL17	Os09t0464300	SORBI_3002G230301	–
13	Zm00001d031826	ZmAHL17	AT5G49700	AHL17	–	SORBI_3007G159800	–
14	Zm00001d042060	–	–	–	Os01t0953801	SORBI_3003G429101	–
15	Zm00001d031887	ZmAHL20	AT4G14465	AHL20	Os08t0563200	–	Pathogen defense ^c [15]
16	Zm00001d002495	–	–	–	Os04t0590200	SORBI_3006G197100	–
17	Zm00001d049521	–	–	–	Os08t0159700	SORBI_3007G050200	–
18	Zm00001d017825	ZmAHL29	AT1G20900	AHL27	Os02t0713700	SORBI_3004G262200	Hypocotyl growth and senescence ^c [3]
19	Zm00001d045928	ZmAHL29	AT1G20900	AHL27	Os06t0326900	SORBI_3010G134900	Hypocotyl growth and senescence ^c [3]
20	Zm00001d046696	ZmAHL29	AT1G20900	AHL27	–	SORBI_3010G196000	Hypocotyl growth and senescence ^c [3]
21	Zm00001d001784	ZmAHL14	AT3G04590	AHL14	Os04t0683900	SORBI_3006G273000	–
22	Zm00001d018515	ZmAHL9	AT2G45850	AHL9	–	–	–
23	Zm00001d051861	ZmAHL9	AT2G45850	AHL9	Os02t0824300	SORBI_3004G352100	–
24	Zm00001d045427	bsf1	–	–	Os06t0136900 ^a	SORBI_3010G027300	Ear develop. ^b [7], Floral develop. ^d [2]
25	Zm00001d006034	ZmAHL10	AT2G33620	AHL10	–	SORBI_3002G249000	Drought tolerance ^c [13]
26	Zm00001d009031	–	–	–	Os01t0246601	–	–
27	Zm00001d039249	–	–	–	–	SORBI_3003G000700	–
28	Zm00001d053429	ZmAHL22	AT2G45430	AHL22	Os02t0448000	SORBI_3004G110700	Hypocotyl growth ^c [4]
29	Zm00001d018508	ZmAHL20	AT4G14465	AHL20	Os02t0820800	SORBI_3004G349125	Pathogen defence ^c [15]
30	Zm00001d028577	ZmAHL15	AT3G55560	AHL15	Os03t0270000	SORBI_3001G423800	Gibberellin homeostasis ^c [18]
31	Zm00001d049906	ZmAHL20	AT4G14465	AHL20	–	SORBI_3007G165400	Pathogen defense ^c [15]
32	Zm00001d026652	ZmAHL14	AT3G04590	AHL14	–	–	–
33	Zm00001d020951	ZmAHL10	AT2G33620	AHL10	–	–	Drought tolerance ^c [13]
34	Zm00001d020768	bsf1	AT5G49700	AHL17	–	–	Ear develop. ^b [7]
35	Zm00001d051620	ZmAHL27	AT1G20900	AHL27	–	–	Hypocotyl growth and senescence ^c [3]
36	Zm00001d026035	–	–	–	–	–	–
37	Zm00001d024537	–	–	–	–	–	–

^a Gene Os06t0136900 is named as *DP1* and none of the other rice are named and the gene name column is omitted.^b Studied in maize.^c Studied in *Arabidopsis*.^d Studied in rice. None of the sorghum *AHL* genes are named hence the gene name column is omitted.

expression (11 AHLs), tissue-specific expression observed only in certain organs (11 AHLs), and stage-specific expression observed only in a subset of developmental stages of certain organs (15 AHLs) (Fig. 5a). The expression for all the genes identified in our study was detected in the transcriptome dataset thus supporting the annotation of maize AHLs.

One of the remarkable features of tissue-specific genes was a complete lack of expression in leaves although the majority of these AHLs (8) showed high expression in all non-leaf organs and the rest (3) had rather restricted expression in cob and immature tassel (Fig. 5a). Surprisingly, all the tissue-specific genes except one (*Zm00001d020951*) belong to the Type-I (Clade-A) category and are, therefore, proposed to be ancestral AHLs [23]. Lack of expression of these AHLs in leaf demonstrates that these genes emerged for specialized function in non-leaf organs. Thus, this subset of Clade-A AHLs might have evolved after the separation of the bryophytes (mosses and liverworts) and the vascular plants and, therefore, functions in the organs specific to the vascular plants. Three tissue-specific genes showing higher expression in cob and immature tassel, which are annotated as *ZmAHL20* (Table 1) and likely paralogs, seem to have a very specific function related to the meiotic activity. Further investigations exploring the roles of various tissue-specific genes will considerably expand our knowledge of both the evolution and the function of AHLs.

The stage-specific genes produced four distinct patterns based on expression in different tissues and organs (Fig. 5a). Two of these clusters (SSC-I and SSC-II) are specific to a very limited number of

developmental stages of a small number of organs and thus represent specific functions. The *barren stalk fastigiate1* (*bsf1*, *Zm00001d045427*), required for the formation of maize ears [7], belongs to SSC-II group and the expression of this gene is detected in cob, tassel, silk, and, surprisingly, internodes. Putative *Arabidopsis* ortholog of another gene (*Zm00001d030053*) is involved in the regulation of hypocotyl growth [4] and, notably, this gene has the highest expression in coleoptile supporting a similar role in maize. The stage-specific genes represent AHLs with higher functional isolation compared to the tissue-specific genes and, thus, regulate very specific processes.

The third set of genes have a constitutive expression in organs and developmental stages and, therefore, perform housekeeping functions. Despite expression in all tissues, this set formed three distinct groups due to high, albeit, distinct expression levels (Fig. 5a). One of the genes in this set (*Zm00001d028577*) is a putative ortholog of *Arabidopsis* *AHL15* involved in maintaining gibberellin homeostasis [18]. Two genes (*Zm00001d031481* and *Zm00001d052901*) are putative orthologs of *Arabidopsis* *AHL10* shown to be involved in drought tolerance [13]. Based on these observations, this set appears to control some of the agronomically important traits warranting functional investigations in maize.

A snapshot of global expression patterns of Type-I, Type-II, and Type-III *AHL* genes showed distinct patterns than those observed in the evolutionary classification (Fig. 5b). Eight genes, including seven Type-I AHLs and one Type-III AHL, lacked expression in all leaf tissues but had expression in all the other tissues and formed a distinct group

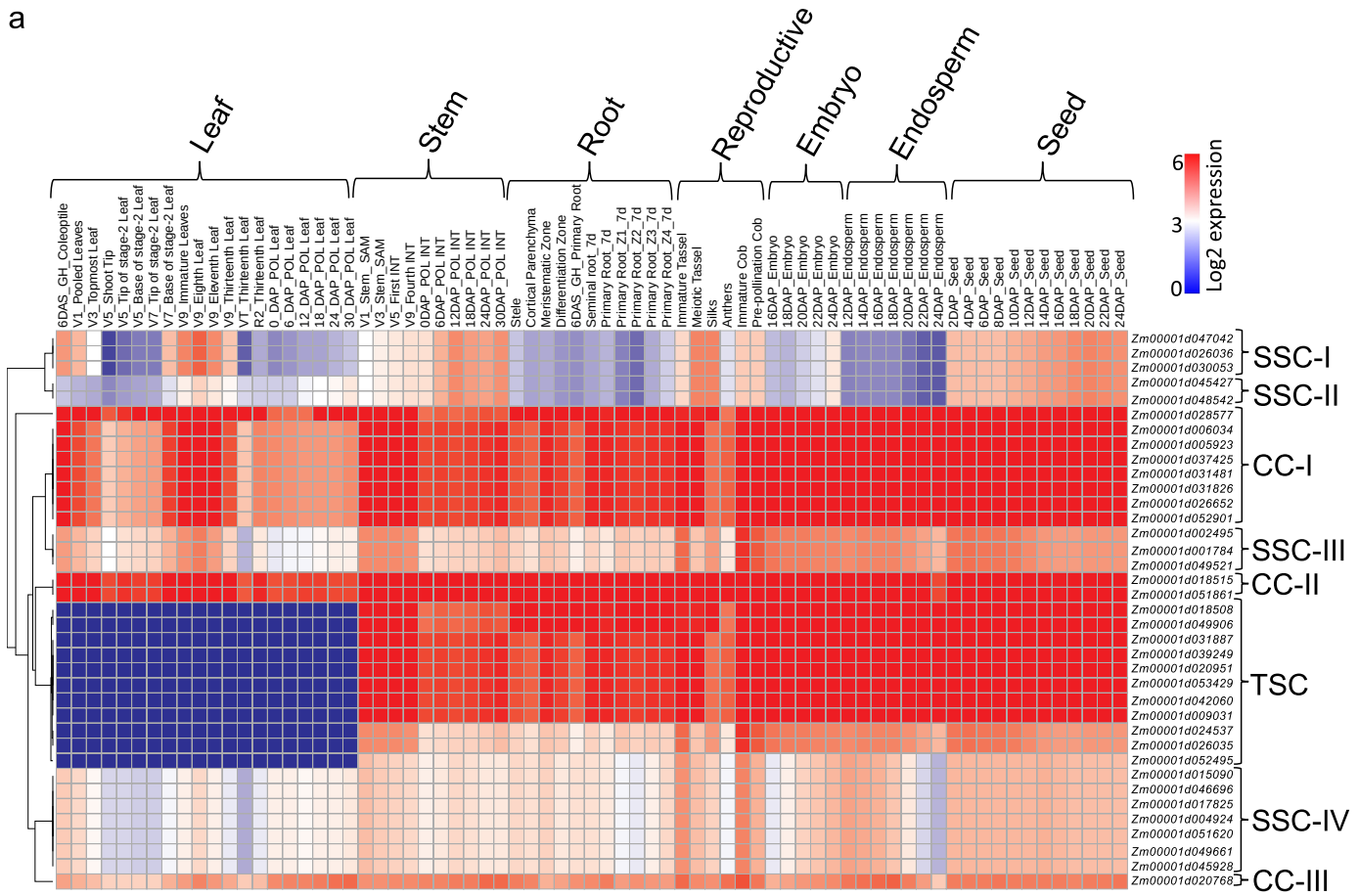


Fig. 5. Clustering of maize AHLs based on gene expression. A) Heatmap showing hierarchical clustering of genes based on gene expression in different tissues and at distinct developmental stages mentioned at the top. Clustering was based on log2-transformed FPKM values. Clusters of genes with stage-specific (SSC), constitutive (CC), and tissue-specific (TSC) expression are identified on the right side. B) PCA plot of all 37 AHLs gene based on gene expression in all tissues shown in A.

(Fig. 5b). Another major cluster of 12 genes included only Type-I and Type-II AHLs designated as stage-specific (SS-I, SS-II, and SS-IV) in the heat map (Fig. 5a–b). These clustering patterns suggest a striking functional divergence of AHLs which could be a result of diversification of the regulatory regions of these genes. Whether this classification of AHLs based on tissue- and stage-specific on gene expression is specific to maize or extends to other grasses will provide useful clues about the specific roles of AHLs in development or functionality of these organs.

3.7. Coexpression networks analysis identified novel genes interacting with AHLs

To identify the genes either regulated by AHLs or acting as co-regulators to modulate the expression of the downstream genes, we performed coexpression analysis. The analysis was performed for all the organ groups discussed above, and coexpression networks were obtained for five organ groups including seed, endosperm, root, leaf, and reproductive tissues. This analysis placed 5 AHLs along with 27 other genes in coexpression networks for these organ groups (Fig. 6, Supplementary Table S4). Four of the networks contain one unique AHLs while one AHL (*Zm00001d052901*) is present in both seed- and leaf-specific networks (Fig. 6).

To test if the genes present in coexpression network together with AHLs can potentially be regulated by AHLs, we searched the promoter regions of these genes for the signature AA(T/A)T sequence bound by AHLs [18]. Remarkably, of the 27 genes coexpressed with AHLs, 25 (93%) carry the signature sequence further supporting their regulation by AHLs. Furthermore, all five AHLs involved in coexpression networks

also carry the signature sequence indicating the regulation of these genes by proteins carrying the AT-hook motif. It should, however, be noted that the AT-hook motif is present in a number of other proteins besides AHLs and, therefore, allow binding of those proteins to DNA carrying the signature sequence.

In the seed-specific network, we obtained one major cluster of 364

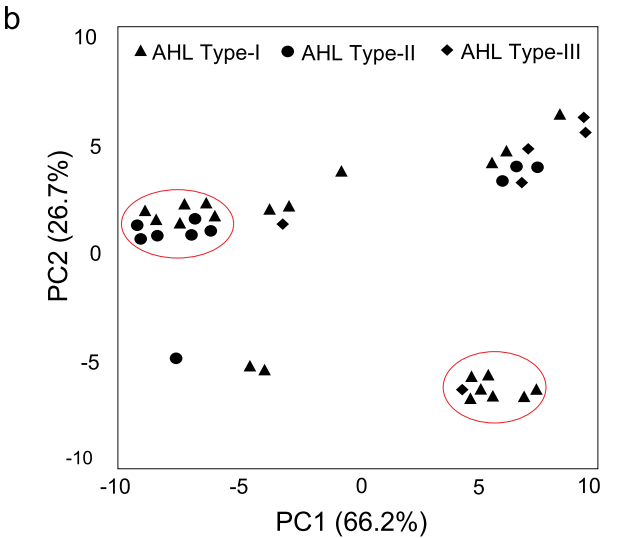


Fig. 5. (continued)

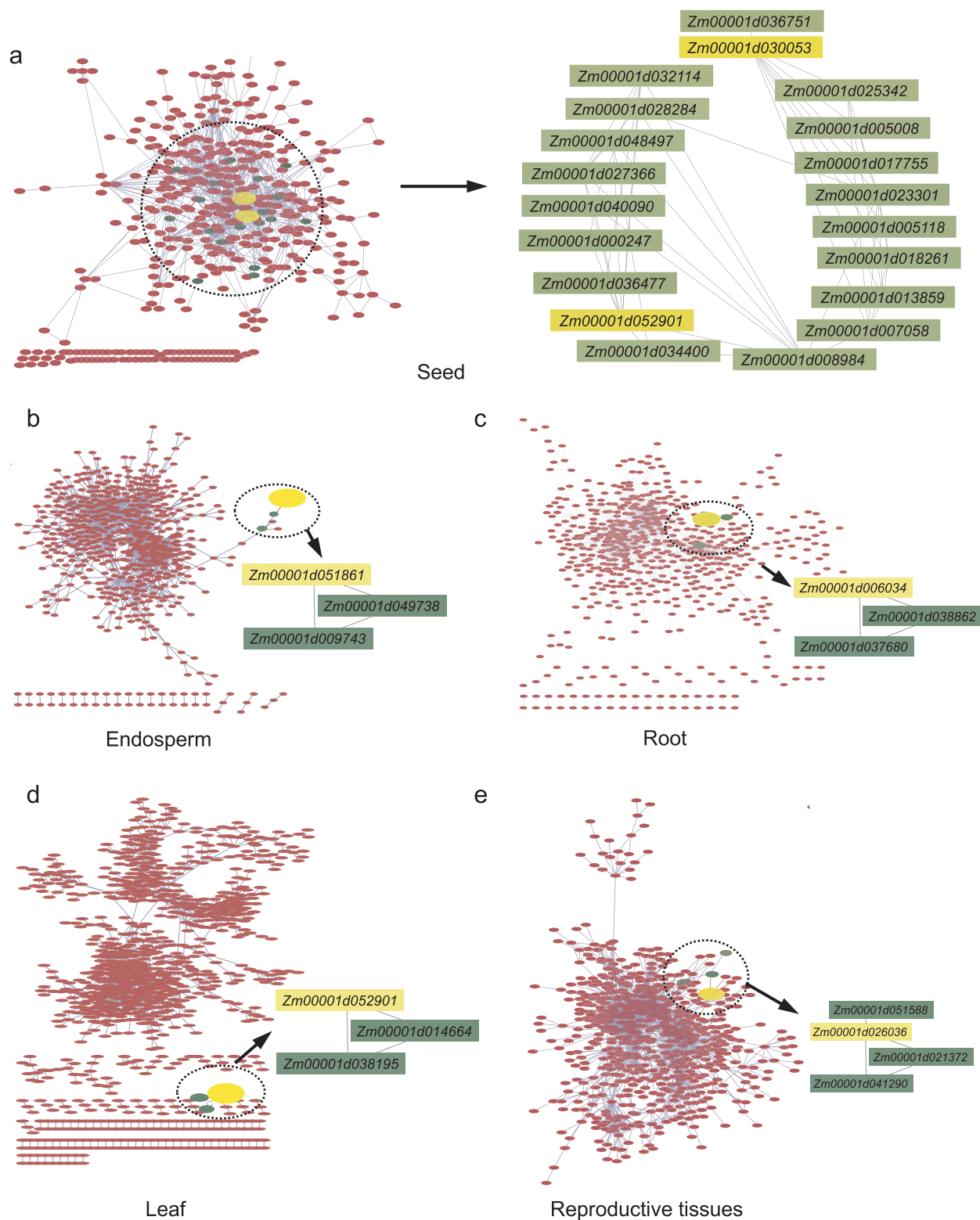


Fig. 6. Co-expression networks involving maize AHLs in five distinct organs. Whole networks for seed (a), endosperm (b), root (c), leaf (d), and reproductive tissues (e) are drawn with edges shown by magenta lines, nodes representing *AHL* gene(s) shown by yellow ovals, genes interacting with the *AHL* gene(s) in the network shown by green ovals, and non-*AHL* interacting genes shown by red ovals. Subnetworks representing the *AHL* gene(s) and their interacting partners for each organ are marked by a dotted oval. Details of these subnetworks including the name of the *AHL* gene(s) (yellow rectangles), name of the interacting non-*AHL* genes (green rectangle), and edges representing the interactions (dark grey lines) are indicated by arrows. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

coexpressing modules providing an elaborate network of genes underlying seed development (Fig. 6a, Supplementary Table S4). Remarkably, two of the nodes in the coexpressing modules are represented by AHL genes, *Zm00001d052901*, and *Zm00001d030053* (Fig. 6a, Supplementary Table S5). Eighteen genes were coexpressed with these two AHLs. None of the genes in this network have been functionally characterized and thus represent novel candidates for understanding seed development. The endosperm-specific network consisted of one major cluster of 575 coexpressing modules, and one of these nodes represented an AHL *Zm00001d051861* (Fig. 6b, Supplementary Table S4). The two genes *Zm00001d049738* and *Zm00001d009743* that were coexpressed with this AHL gene (Fig. 6; Supplementary Table S5). *Zm00001d049738* is a putative ortholog of *Transmembrane protein 18* (*TMEM18*), a pentatricopeptide repeat-containing protein (PPR), which regulates pollen tube and vegetative growth in *Arabidopsis* [67]. In the root-specific network, we obtained one major cluster of 531 coexpressing modules and one of the nodes represented AHL *Zm00001d006034* (Fig. 6c, Supplementary Table S4). The two other genes, *Zm00001d038862* and *Zm00001d037680*, were coexpressed with the AHL gene (Fig. 6c, Supplementary Table S5). Remarkably, *AHL10*, a putative *Arabidopsis* ortholog of *Zm00001d006034*, has been shown to be involved in drought tolerance by controlling root elongation [13]. Therefore, *Zm00001d006034* represents a valuable target for the improvement of drought tolerance. One of the coexpressed gene, *Zm00001d038862*, is a putative ortholog of *Arabidopsis LOG8* which regulates senescence by activation of cytokinin [68]. The other gene, *Zm00001d037680*, is a putative ortholog of *Arabidopsis Vsp2* involved in wound response [69]. Based on these putative functions, the genes in this network appear to have an important role in root development that needs to be investigated. In leaf, we obtained one cluster of 3 coexpressing modules and one of the nodes consisted of an AHL gene *Zm00001d052901* (Fig. 6d, Supplementary Table S4). Two genes, *Zm00001d038195* and *Zm00001d014664*, were coexpressed with the AHL gene (Fig. 6d; Supplementary Table S5). Interestingly, *Zm00001d052901* is an ortholog of *Arabidopsis AHL10* that is also present in the root-specific network and involved in drought tolerance [13]. Support for the role of this network in leaf development is provided by the observation that *CIA2*, a putative *Arabidopsis* ortholog of *Zm00001d014664*, is involved in importing proteins from the cytoplasm to chloroplasts [70]. Such import is essential for maintaining the normal photosynthesis in chloroplast and lack thereof leads to cell death and senescence. In the reproductive tissue, we obtained one major cluster of 649 coexpressing modules with one node representing an AHL gene *Zm00001d026036* (Fig. 6e; Supplementary Table S5). Three genes (*Zm00001d051588*, *Zm00001d021372*, and *Zm00001d041290*) were coexpressed with this AHL gene (Fig. 6e; Supplementary Table S4). All these genes lack functional characterization and represent novel candidates.

3.8. Protein networks provide insights about AHL interacting proteins

The AHL proteins catalyze various biological activities by binding with DNA and are, therefore, expected to form complexes with other proteins to carry out such regulatory roles. We constructed three protein interaction networks consisting of 8 AHLs and 4 interacting proteins (Fig. 7). The interacting proteins included DNA repair protein RAD1 (*Zm00001d051861*) (Fig. 7a), pentatricopeptide repeat-containing protein PRCP (*Zm00001d033731*) (Fig. 7b and c), and heat shock transcription factor-like protein HSP29 (*Zm00001d016520*) (Fig. 7c). Consistent with the predicted interactions, the AHLs are shown to interact with each other and to other proteins through their PPC domain [4]. However, the predicted interactions involving the proteins other than AHL are novel and needs functional characterizations to reveal their role in maize development.

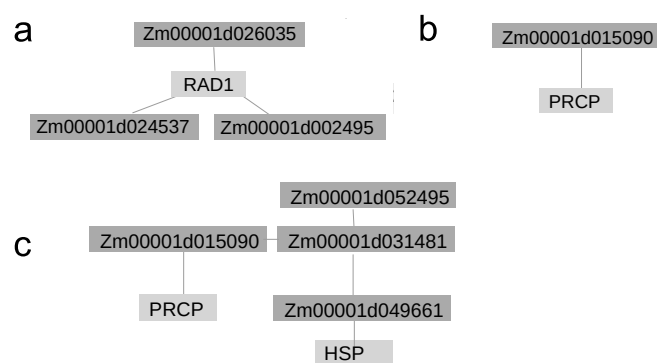


Fig. 7. Protein interaction networks involving maize AHLs. The AHL are in dark grey rectangles and their interacting partners are shown in light grey rectangles.

4. Conclusions

Importance of the AHL gene family in economically important traits including plant defense and drought tolerance is evident from studies in *Arabidopsis*. A striking lack of knowledge about the role of this family in crop plants underscores the importance of functional characterization of this gene family. Crop productivity is dependent on the ability of plants to withstand biotic and abiotic stresses and, therefore, functional characterization and implication of AHLs genes in maize, one of the most important food crops in the world, holds great potential in crop improvement. AHLs are also involved in some of the fundamental developmental processes and, therefore, the aforementioned knowledge will help in further understanding the underlying biology of plants. Maize genome also carries a large number of transposable elements that are generally silenced via epigenetic mechanisms to preserve genome integrity and function. Given the emerging role of AHLs in the regulation of epigenetic states, some of the genes reported in our study are promising candidates orchestrating such regulation. Besides identification and characterization of AHLs in maize, identification of new AHLs in rice and sorghum will help in further understanding the role of this gene family in these important cereal crops. Finally, the identification of the genes that putatively interact with AHLs at the transcriptional and posttranscriptional level will be valuable for a comprehensive understanding of the pathways regulated by the AHL gene family.

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Authors' contributions

RS, EB, and RK designed this study. EB, RK, FL, and CS performed computational and data analyses and analyzed the data. RS, EB, and RK wrote the manuscript. All authors provided input on manuscript draft.

Declaration of Competing Interest

The authors declare that they have no competing interests.

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